Screening for prostate cancer--how to manage in 2006?

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National Societies usually recommend screening for Prostate Cancer (PC) with Serum Prostate Specific Antigen (PSA) and digital rectal examination annually beginning at age 50. In high risk population including men with a family history of PC or African population screening should start at age of 45 years. PSA has been widely used to detect PC despite the fact that PSA is not specific for PC. Over the years serum PSA level of greater than 4.0 ng/ml was considered the treshold to perform prostate biopsy, searching for PC. In 2005 the Prostate Cancer Prevention Trial (PCPT) demonstrated that the cut-off of 4.0 ng/ml for PSA is not anymore adapted due to the fact that this survey found in 15% of men with PSA < or = 4.0 ng/ml a prostate cancer on sextant biopsies. Today the value of PSA and the cut-off for Prostate biopsy is questionned suggesting that PSA level higher than 2.6 ng/ml must be the case to propose Prostate Biopsy. Catalona confirms that approximately 25% to 30% of men with PSA 2.6 to 4.0 ng/ml have prostate cancer. Schröder and Gosselaar assert that screening for PC at low PSA levels (< 4.0 ng/ml) risks to detect clinically insignificant cancers which are no threat to man. So far in the year 2006 screening for PC demonstrates accumulating evidences of efficacy but persistent uncertainty. The major question for an urologist at work when facing a young men searching early diagnosis of PC is: at which level of PSA do we have to perform rectal biopsy?

[Prostatic carcinoma. Screening--when and what?].

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Prostate cancer is now the most common cancer and the second most common cause of death from cancer among men. Therapy of curative intention is only possible in organ confined disease. The use of prostate specific antigen (PSA) and digital rectal examination (DRE) results in a three fold increase in prostatic carcinoma detection. Levels of PSA > 4 ng/ml are indications for sextant biopsies of the prostate. There did not exist an intermediate range or 'grey zone' of PSA 4-10 ng/ml where wait and see diagnostic procedure is indicated. In PSA levels > 10 ng/ml curative therapy can only performed in 15-44% of the cases. PSA and DRE examination should be performed between the age of 50 and 70 years when life expectancy exceeds ten years. In case of familiar history the case finding has to start at the age of 45. There is no support for the common opinion that early detection finds clinically insignificant cancer since autoptical prevalence of prostate cancer is about 40% and early detection discover only 3-4%. Results about the usefulness of active screening in a population will be available in 2005.

Prostate Cancer Screening: Common Questions and Answers.

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Prostate cancer is the most diagnosed noncutaneous malignancy and the second most common cause of cancer death among men in the United States. Risk factors include older age, family history of prostate cancer, and Black race. Screening via prostate-specific antigen testing may lead to a small reduction in prostate cancer-specific mortality, with no reduction in all-cause mortality, but it can cause significant harms related to false-positive test results, unnecessary biopsies, overdiagnosis, and overtreatment. Shared decision-making is strongly recommended by all national guidelines before initiating screening.

Most guidelines recommend screening every 2 to 4 years in men 55 to 69 years of age at average risk. After a positive prostate-specific antigen test result (more than 4 ng/mL), the test should be repeated. If the prostate-specific antigen level is still elevated, next steps include multiparametric magnetic resonance imaging, assessment of urine or blood biomarkers, and referral to urology. Active surveillance is increasingly accepted as the preferred standard of care for patients with newly diagnosed low-risk prostate cancer, because it is associated with similar long-term survival and better quality of life than curative treatment. The primary intent of screening is to identify patients with clinically significant prostate cancer who may benefit from curative treatment while minimizing the detection of clinically insignificant cancer.

Viewpoint: limiting prostate cancer screening.

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Prostate cancer screening is controversial, and major professional associations offer differing screening guidelines. The author addresses 3 key issues about prostate cancer screening: 1) the prostate-specific antigen (PSA) criteria to recommend a prostate biopsy, 2) the appropriate age to start screening, and 3) the appropriate age to stop screening. The author argues, on the basis of evidence published since 2000, that data supporting the efficacy of PSA screening remain unconvincing. The author recommends that screening should not be expanded to include average-risk men younger than age 50 years or older than age 75 years and that a PSA threshold below 4.0 ng/mL should not be used to trigger biopsy referral.

Results of compliance with prostate cancer screening guidelines.

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Current guidelines of the American Cancer Society and the National Comprehensive Cancer Network recommend offering annual prostate cancer screening with prostate specific antigen (PSA) and digital rectal examination (DRE) beginning at age 50 (age 45 in high risk men). There are limited data concerning outcomes if all men followed screening guidelines. We report early outcome data on men who entered a prostate cancer screening study, complied with the screening guidelines and were subsequently diagnosed with prostate cancer. We reviewed records of men 45 to 59 years old at study entry with a PSA less than 2.6 ng/ml and benign DRE who underwent annual DRE and PSA testing in a screening study between 1991 and 2001. Of 10,174 men with these characteristics, 232 (2.3%) were subsequently diagnosed with prostate cancer. We evaluated PSA, Gleason score, clinical and pathological tumor stage, and treatment outcomes in these men. Median PSA at diagnosis was 3.1 ng/ml (range 0.4 to 9.6). Gleason scores ranged from 4 to 9. All patients had clinically localized disease. Management included predominantly radical prostatectomy (87%) and radiation therapy (10%). Of cancers in which tumor volume was assessed 13% were considered possibly harmless tumors by previously published criteria and 2% were considered possibly rapidly progressive tumors by criteria we set in this study. Prostate cancer screening using some current guidelines results in the detection of cancers that are organ confined in 79% of patients, possibly harmless in less than 15% and possibly rapidly progressive in 2%.

Men's Health: Prostate Cancer Screening.

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Prostate cancer is the second most common nondermatologic cancer in males in the United States. The median age at diagnosis is 66 years and median age at death is 80 years, with most patients diagnosed between ages 55 and 74 years. Black men are at greatest risk of developing and dying of prostate cancer. The U.S. Preventive Services Task Force (USPSTF) and American Urological Association (AUA) guidelines recommend shared decision-making in consideration of screening for men ages 55 to 69 years. Currently, digital rectal examination alone is not recommended for prostate cancer screening. The serum prostate-specific antigen (PSA) test remains the most common screening tool. Novel formulas and algorithms, including the Prostate Health Index (phi) and the 4Kscore, which use total PSA, free PSA, and other information to estimate risk, have shown greater predictive values for detection than the PSA test. Risk assessment with magnetic resonance imaging (MRI) study with or without MRI/transrectal ultrasonography (TRUS) targeted biopsy requires fewer biopsy specimens than traditional TRUS-guided biopsy, and is associated with higher detection rates. Studies of specific lifestyle modifications to minimize prostate cancer risk have shown inconclusive results; however, high carbohydrate and animal fat intakes may increase the risk.

The New US Preventive Services Task Force "C" Draft Recommendation for Prostate Cancer Screening.

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The US Preventive Services Task Force has issued a new draft guideline, with a "C" recommendation that men aged 55-69 yr should be informed about the benefits and harms of screening for prostate cancer, and offered prostate-specific antigen testing if they choose it. For men aged ≥70 yr, the recommendation remains "D", or "do not screen." This draft represents substantial progress in the right direction towards offering men a fair opportunity to discuss the risks and benefits of screening with their primary care providers. However, the evidence review underlying the draft remains fundamentally inadequate, leading to biased presentations of both benefits and harms of screening. The final guideline and future revisions should reflect formal engagement with subject matter experts to optimize the advise given to men and their physicians.

Shared Decision Making Emphasized for Prostate Screening.

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The U.S. Preventive Services Task Force (USPSTF) now recommends that men ages 55 to 69 individually decide, in consultation with their physician, whether to undergo PSA screening for prostate cancer. The guidelines represent a change from 2012, when the USPSTF recommended against screening for all men.

Early detection of prostate cancer: AUA Guideline.

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The guideline purpose is to provide the urologist with a framework for the early detection of prostate cancer in asymptomatic average risk men. A systematic review was conducted and summarized evidence derived from over 300 studies that addressed the predefined outcomes of interest (prostate cancer incidence/mortality, quality of life, diagnostic accuracy and harms of testing). In addition to the quality of evidence, the panel considered values and preferences expressed in a clinical setting (patient-physician dyad) rather than having a public health perspective. Guideline statements were organized by age group in years (age <40; 40 to 54; 55 to 69; ≥ 70). Except prostate specific antigen-based prostate cancer screening, there was minimal evidence to assess the outcomes of interest for other tests. The quality of evidence for the benefits of screening was moderate, and evidence for harm was high for men age 55 to 69 years. For men outside this age range, evidence was lacking for benefit, but the harms of screening, including over diagnosis and overtreatment, remained. Modeled data suggested that a screening interval of two years or more may be preferred to reduce the harms of screening. The Panel recommended shared decision-making for men age 55 to 69 years considering PSA-based screening, a target age group for whom benefits may outweigh harms. Outside this age range, PSA-based screening as a routine could not be recommended based on the available evidence.

Reconciling primary care and specialist perspectives on prostate cancer screening.

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When specialists propose screening guidelines for primary care clinicians to implement, differences in perspectives between the 2 groups can create conflicts. Two recent specialty organization guidelines illustrate this issue. The American Urological Association guideline panel and National Comprehensive Cancer Network recommend that average-risk men first be counseled about the risks and benefits of prostate-specific antigen screening for prostate cancer at age 40 rather than at the previously recommended age of 50 years. There is no direct evidence, however, that this recommendation has any impact on prostate cancer-specific mortality. To avoid distracting primary care clinicians from providing services with proven benefit and value for patients, professional organizations should follow appropriate standards for developing guidelines. Primary care societies and health care systems should also be encouraged to evaluate the evidence and decide whether implementing the recommendations are feasible and appropriate.